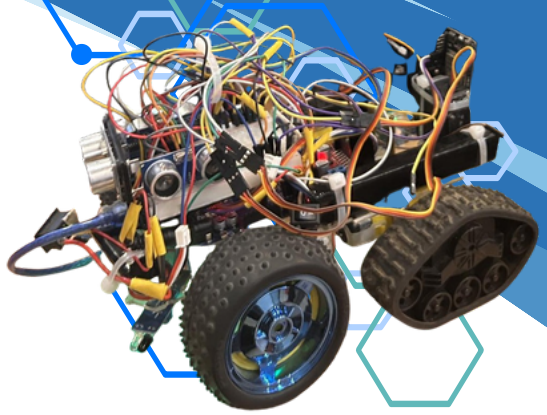
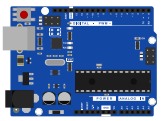
carNaps

A sensor-based search and rescue robotic system using ultrasonic and infrared detection

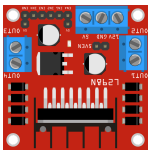


carNaps is an autonomous robotic system designed to detect obstacles and collect objects using multiple sensors and a motorized claw mechanism. The robot utilizes three ultrasonic sensors to measure distance and identify obstacles in its surroundings, while two infrared sensors assist in identifying diagonal proximity and object detection.

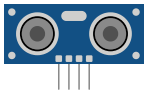
Hardware Components



1x Arduino R3 UNO



1x L298 Motor Driver

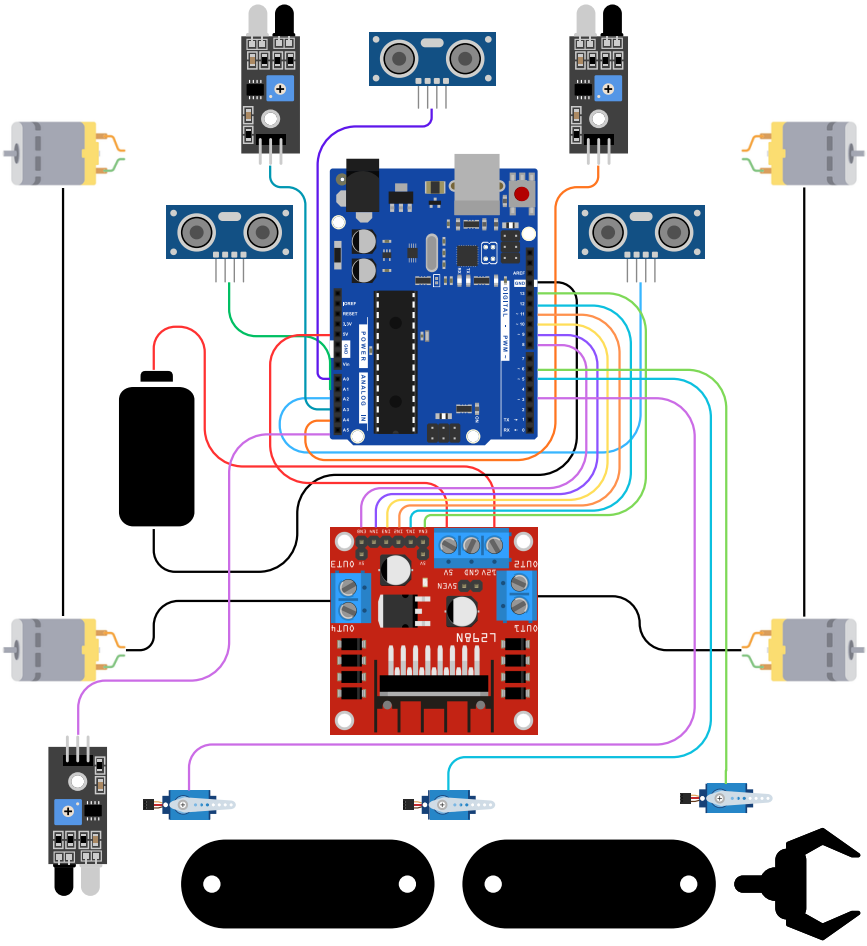


3x Ultrasonic Sensors

3x IR Proximity Sensors

4x DC Motors

3x Servo Motors



Key Features

- Multi-sensor obstacle detection
- Autonomous navigation
- Robotic claw manipulation
- Efficient motor control
- Integrated sensor processing

Applications

- Search and rescue operations
- Autonomous delivery
- Warehouse object collection
- Robotics education and research